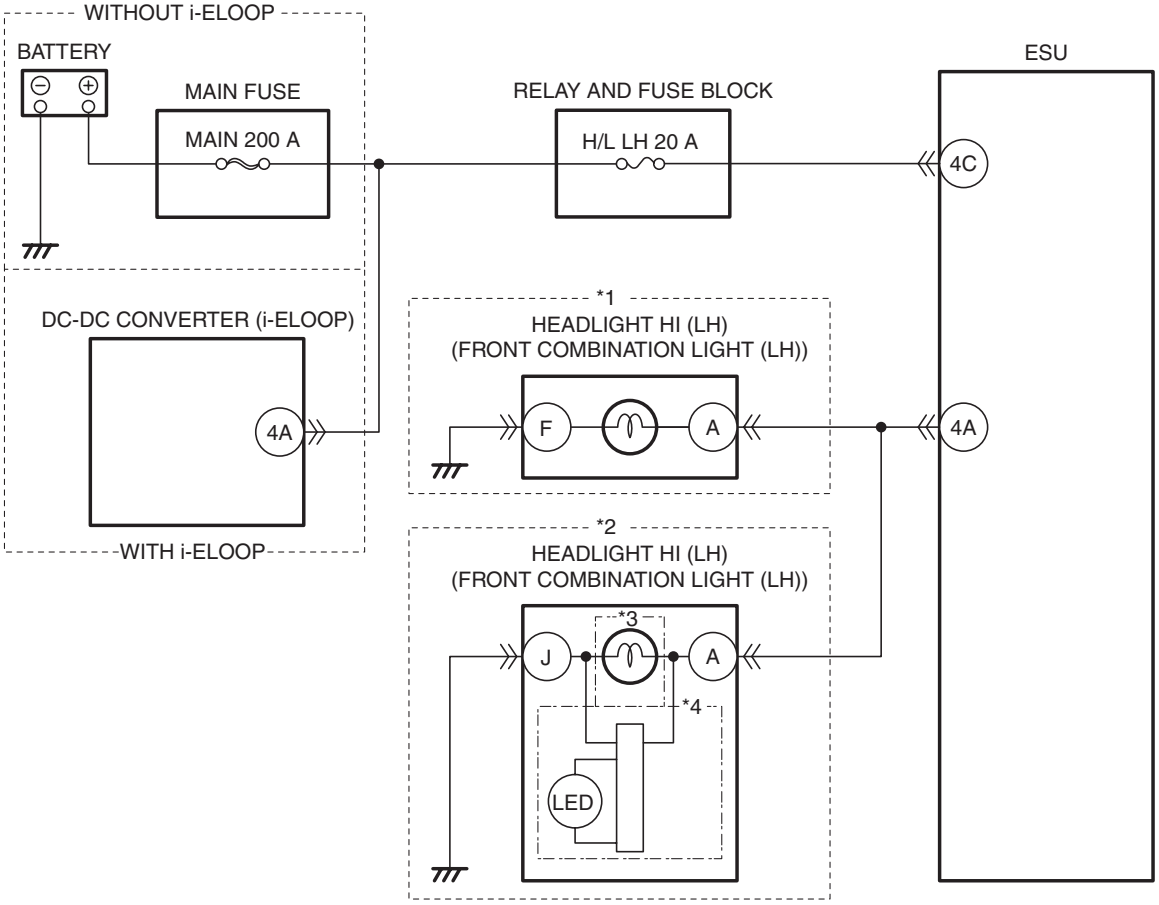
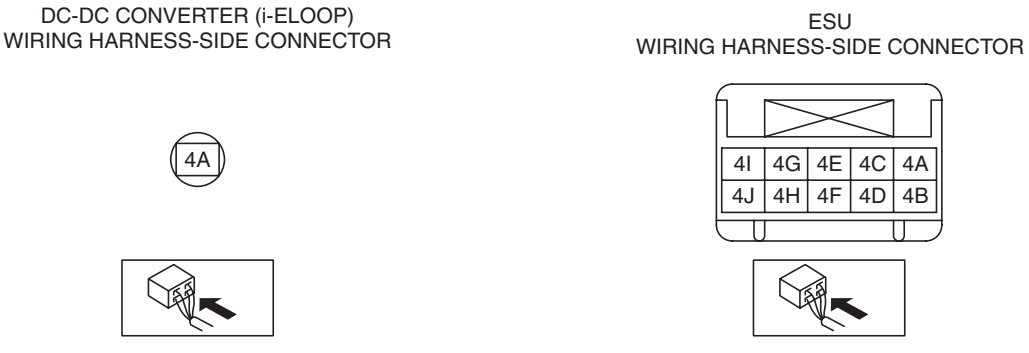
DTC B1D02:13 [ELECTRICAL SUPPLY UNIT (ESU)]

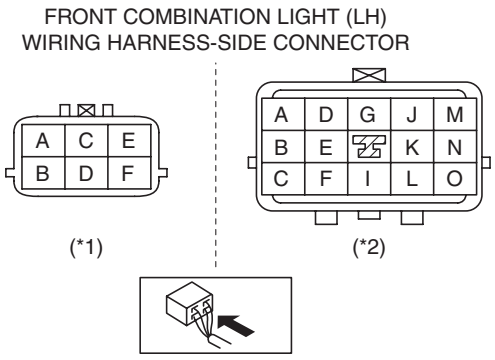
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Description	Headlight HI (LH) circuit malfunction
Detection condition	• Electrical supply unit (ESU) detects an open circuit in the headlight HI (LH) circuit.
Fail-safe	Not applicable
Possible cause	• Electrical supply unit (ESU) connector or terminal malfunction • Electrical supply unit (ESU) power supply circuit malfunction — Without i-ELOOP: • Short to ground in wiring harness between MAIN 200 A fuse and electrical supply unit (ESU) terminal 4C • H/L LH 20 A fuse malfunction • MAIN 200 A fuse malfunction • Open circuit in wiring harness between battery positive terminal and electrical supply unit (ESU) terminal 4C — With i-ELOOP: • Short to ground in wiring harness between H/L LH 20 A fuse and electrical supply unit (ESU) terminal 4C • H/L LH 20 A fuse malfunction • Open circuit in wiring harness between DC-DC converter (i-ELOOP) terminal 4A and electrical supply unit (ESU) terminal 4C • Headlight HI/LO bulb (LH) malfunction • Front combination light (LH) connector or terminal malfunction • Open circuit in wiring harness between the following terminals: — Front combination light (LH) terminal F—Body ground (With front combination light (bulb type) and without headlight manual leveling system) — Front combination light (LH) terminal J—Body ground (With front combination light (LED type) or headlight manual leveling system) • Open circuit in wiring harness between front combination light (LH) terminal A and electrical supply unit (ESU) terminal 4A • Front combination light (LH) malfunction • Electrical supply unit (ESU) malfunction

Description	Headlight HI (LH) circuit malfunction







*1 : With front combination light (bulb type) and without headlight manual leveling system

*2 : With front combination light (LED type) or headlight manual leveling system

*3 : With front combination light (bulb type)

*4 : With front combination light (LED type)

Diagnostic Procedure

Step	Inspection		Action
1	**INSPECT ELECTRICAL SUPPLY UNIT (ESU) CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the electrical supply unit (ESU) connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	**Without i-ELOOP:** • Go to the next step. **With i-ELOOP:** • Go to Step 3.
		No	Repair or replace the connector, then go to Step 9.
2	**VERIFY ELECTRICAL SUPPLY UNIT (ESU) POWER SUPPLY VOLTAGE** • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Measure the voltage at the electrical supply unit (ESU) terminal 4C (wiring harness-side). • Is the voltage **B+**?	Yes	**With front combination light (bulb type):** • Go to Step 4. **With front combination light (LED type):** • Go to Step 5.
		No	Inspect the H/L LH 20 A fuse and MAIN 200 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between MAIN 200 A fuse and electrical supply unit (ESU) terminal 4C. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and electrical supply unit (ESU) terminal 4C. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 9.

Step	Inspection		Action
3	**VERIFY ELECTRICAL SUPPLY UNIT (ESU) POWER SUPPLY VOLTAGE** • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Measure the voltage at the electrical supply unit (ESU) terminal 4C (wiring harness-side). • Is the voltage **B+**?	Yes	**With front combination light (bulb type):** • Go to the next step. **With front combination light (LED type):** • Go to Step 5.
		No	Disconnect the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) Inspect the H/L LH 20 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between H/L LH 20 A fuse and electrical supply unit (ESU) terminal 4C. **If there is a common connector:** • Determine the malfunctioning location by inspecting the common connector and terminals for corrosion, damage, or disconnection and by inspecting the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground, or the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter (i-ELOOP) terminal 4A and electrical supply unit (ESU) terminal 4C. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Connect the service plug, then go to Step 9. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].)
4	**VISUALLY INSPECT HEADLIGHT HI/LO BULB (LH)** • Visually inspect the headlight HI/LO bulb (LH). • Is the headlight HI/LO bulb (LH) normal?	Yes	Go to the next step.
		No	Replace the headlight HI/LO bulb (LH), then go to Step 9. (See HEADLIGHT BULB REMOVAL/INSTALLATION.)
5	**INSPECT FRONT COMBINATION LIGHT (LH) CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the front combination light (LH) connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 9.

Step	Inspection		Action
6	**INSPECT HEADLIGHT HI (LH) GROUND CIRCUIT FOR OPEN CIRCUIT** • Verify that the front combination light (LH) connector is disconnected. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — Front combination light (LH) terminal F (With front combination light (bulb type) and without headlight manual leveling system) — Front combination light (LH) terminal J (With front combination light (LED type) or headlight manual leveling system) • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Front combination light (LH) terminal F—Body ground (With front combination light (bulb type) and without headlight manual leveling system) • Front combination light (LH) terminal J—Body ground (With front combination light (LED type) or headlight manual leveling system) **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 9.
7	**INSPECT HEADLIGHT HI (LH) CIRCUIT FOR OPEN CIRCUIT** • Verify that the front combination light (LH) and electrical supply unit (ESU) connectors are disconnected. • Inspect the wiring harness for continuity between front combination light (LH) terminal A (wiring harness-side) and electrical supply unit (ESU) terminal 4A (wiring harness-side). • Is there continuity?	Yes	**With front combination light (bulb type):** • Go to Step 9. **With front combination light (LED type):** • Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between front combination light (LH) terminal A and electrical supply unit (ESU) terminal 4A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step9.
8	**VERIFY IF MALFUNCTIONING LOCATION IS FRONT COMBINATION LIGHT (LH) DEPENDING ON REPEATABILITY** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the electrical supply unit (ESU) using the M-MDS. (See CLEARING DTC [ELECTRICAL SUPPLY UNIT (ESU)].) • Operate the light switch to the HEAD position and the dimmer switch to HI position. • Retrieve the electrical supply unit (ESU) DTCs using the M-MDS. (See DTC INSPECTION [ELECTRICAL SUPPLY UNIT (ESU)].) • Is the same DTC displayed?	Yes	Replace the front combination light (LH), then go to the next step. (See FRONT COMBINATION LIGHT REMOVAL/ INSTALLATION.)
		No	Go to Step 10.

Step	Inspection		Action
9	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the electrical supply unit (ESU) using the M-MDS. (See CLEARING DTC [ELECTRICAL SUPPLY UNIT (ESU)].) • Operate the light switch to the HEAD position and the dimmer switch to HI position. • Retrieve the electrical supply unit (ESU) DTCs using the M-MDS. (See DTC INSPECTION [ELECTRICAL SUPPLY UNIT (ESU)].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the electrical supply unit (ESU). (See ELECTRICAL SUPPLY UNIT (ESU) REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
10	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [ELECTRICAL SUPPLY UNIT (ESU)].)
		No	DTC troubleshooting completed.